**ST MARY'S DSG, KLOOF**

**MATHEMATICS PAPER 2 (Edited)**

**GRADE 10** **19 NOVEMBER 2019**

**TIME: 2 HOURS** **TOTAL: 112 MARKS**

**NAME:**_____ **MATHS TEACHER:**_____

**READ THE FOLLOWING INSTRUCTIONS CAREFULLY:**

1. This question paper consists of 17 pages and 17 questions.
2. Formulae have been provided below.
3. There are 4 sections. Write your name and your Mathematics teacher's name on each section and hand in separately.
4. Answer all the questions.
5. Diagrams are not necessarily drawn to scale.
6. **All the necessary working details must be clearly shown.**
7. **Reasons must be given unless specified otherwise.**
8. Approved non-programmable calculators may be used except where otherwise stated. Ensure that your calculator is in degree mode.
9. It is in your own interest to write legibly and to present your work neatly.
10. Answers may be left in surd format or rounded off to **two decimal places**.

**FORMULAE**

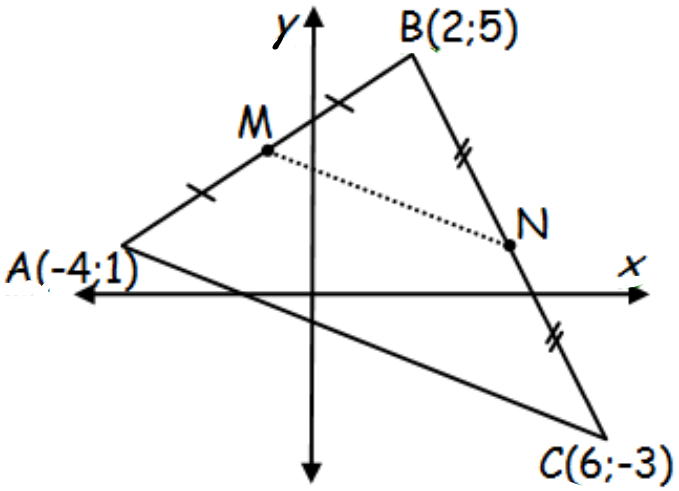
$y = mx + c$ $\quad m = \frac{y_2 - y_1}{x_2 - x_1}$ $\quad d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ $\quad m = \tan\theta$

$M = \left(\frac{x_1 + x_2}{2} ; \frac{y_1 + y_2}{2}\right)$

$A = \pi r^2$ $\quad V = Area\ of\ base \times Height$

**Question 1**

In the diagram below, A(-4;1), B(2;5) and C(6;-3) are the coordinates of ΔABC.
M and N are the midpoints of AB and BC respectively.




a) Give the co-ordinates of M (3)

b) Determine the equation of the straight line BC (4)

c) Calculate the length of AC (3)

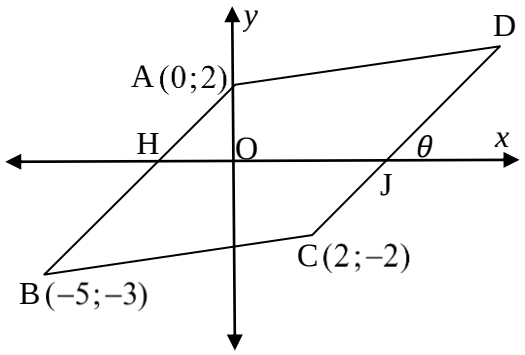
d) Explain clearly why $B\hat{M}N = B\hat{A}C$ . (3)

[13]

**Question 2**

A$(0;2)$, B$(-5;-3)$, C$(2;-2)$ and D are the vertices of parallelogram ABCD.

The *x*-intercept of AB is point H.
CD cuts the *x*-axis at J and makes an angle of $\theta$, as indicated, with the *x*-axis.




a) Determine the gradient of $AB$. (2)

b) Calculate the size of $\theta$ correct to one decimal place. (3)

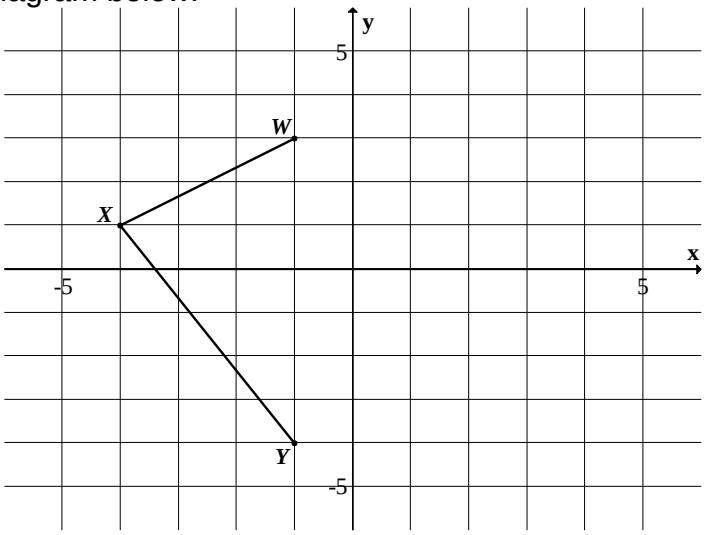
c) Determine the value of the x-intercept at H. (3)

d) Determine the coordinates of D. (2)

[10]

**Question 3**

a) Refer to the diagram below.

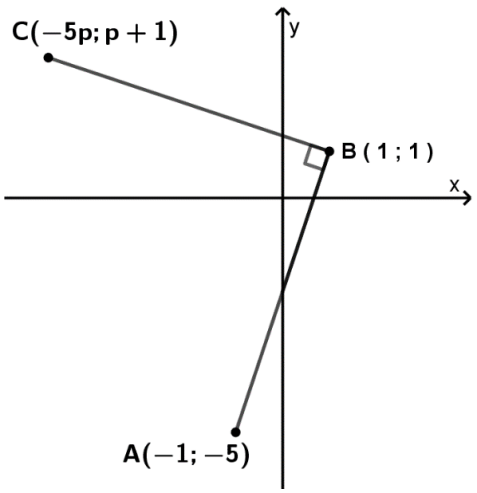



Give the co-ordinates of the point *Z*, so that WXYZ is a kite. _____ (2)

b) In the diagram alongside, which is not drawn to scale, $AB \perp BC$.

Points $A(-1;-5)$, $B(1;1)$ and $C(-5p;p+1)$ are shown.

Determine the value of *p*.




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(5)

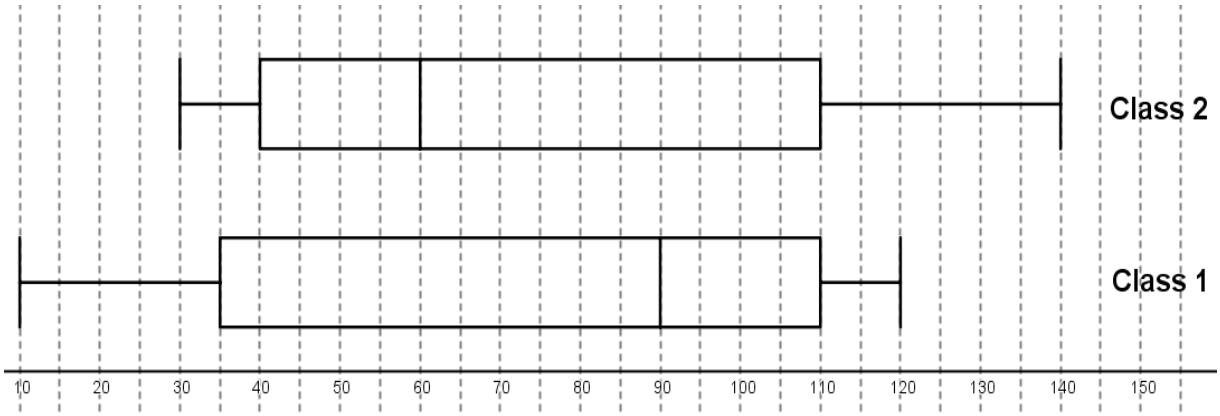
[7]

**SECTION B**

**NAME:**_____ **MATHS TEACHER:**_____

**Question 5**

The exam results of 2 Maths classes were recorded and shown in the box and whisker diagrams below. The exam total is 150 marks.




State whether the following statements are <u>True or False</u>.

a) The range of the classes are equal. _____ (1)

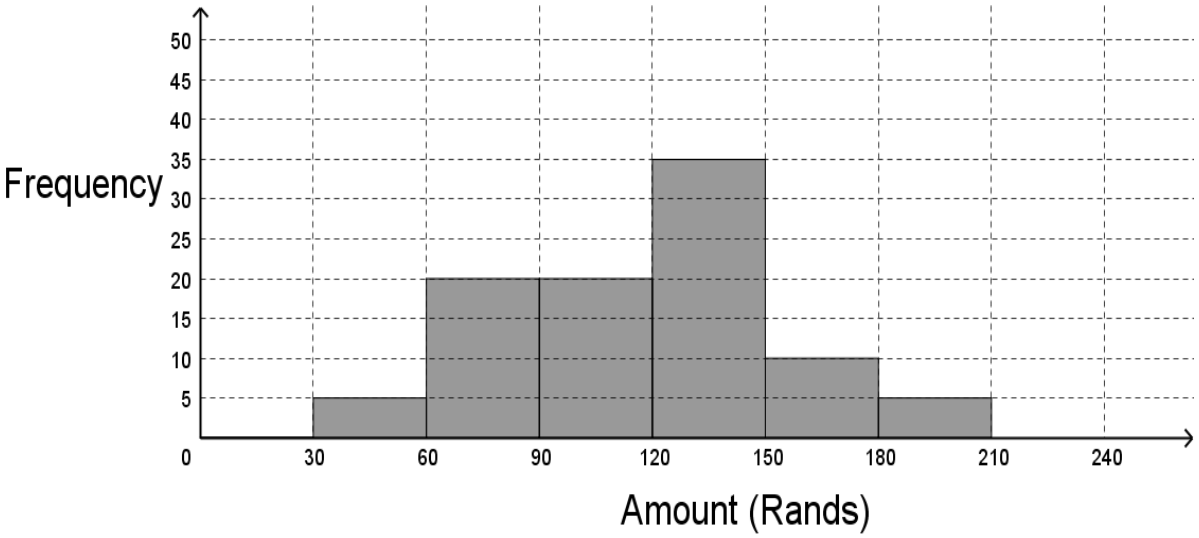
b) Most results in class 1 lie between 35 and 90. _____ (1)

c) The interquartile range for class 2 is 70. _____ (1)

[3]

**Question 6**

The Histogram below shows the amount of money, in Rands, that a Grade 10 group raised at their Entrepreneur Day.



a) Complete the table below from the data given in the Histogram. (1)

| Amount ($x$) in Rands | Frequency |
|---|---|
| $30 < x \leq 60$ | 5 |
| $60 < x \leq 90$ | 20 |
| $90 < x \leq 120$ | |
| $120 < x \leq 150$ | |
| $150 < x \leq 180$ | |
| $180 < x \leq 210$ | |
| **Total:** | **95** |

b) Use a coloured pen to draw a frequency polygon over the Histogram. (2)

c) Determine the modal interval. _____ (1)

d) Determine into which interval the lower quartile $(Q_1)$ falls. _____ (2)

e) Calculate the approximate mean amount of money raised by a Grade 10 pupil. Show all working details.

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(4)

[10]

**Question 7**

Six data values are listed in ascending order:

$$a \; ; \; b \; ; \; 7 \; ; \; c \; ; \; 11 \; ; \; 17$$

Determine the values of $a, b$ and $c$, if the following further information is given regarding the data: (The mean value is represented by $\bar{x}$)

The range $= 13$. $\bar{x} = \frac{28}{3}$ and $Q_2 = 8,5$

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[6]

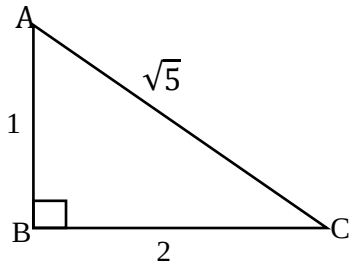
**Question 8**

a) Use your calculator to evaluate correct to 2 decimal places:

1) $\cos 66,6^o + 1$ _____ (1)

2) $\sin^2 20^o$ _____ (1)

b) Use the diagram below to complete the following:

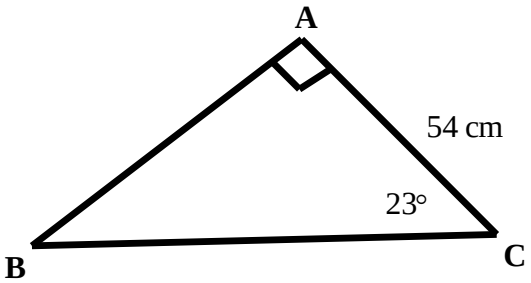



1) cos C = _____ (1)

2) sin A = _____ (1)

3) _____ A = 2 (1)

c) Determine the length of BC, correct to two decimal places. (3)




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[8]

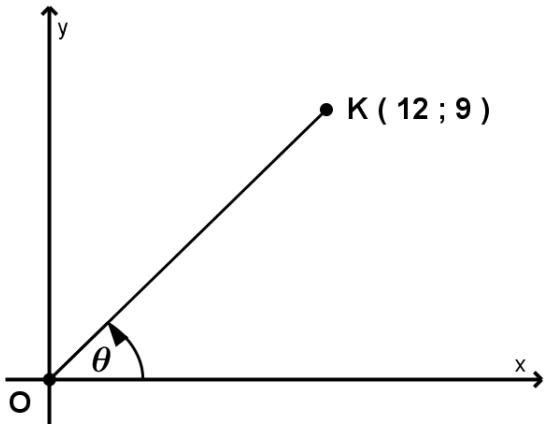
**SECTION C**

**NAME:**_____ **MATHS TEACHER:**_____

**Question 9**

$K$ is the point $(12;9)$ and $K\hat{O}X = \theta$

Use the diagram to answer the questions that follow:




a) Determine, without the use of a calculator:

1) $\tan\theta$ (1)

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2) $3\cos\theta$ (3)

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3) the value of $m$ if $\sin\theta = \frac{2}{m}$ (3)

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b) Use a calculator to find the value of $\theta$ in the above diagram. (2)

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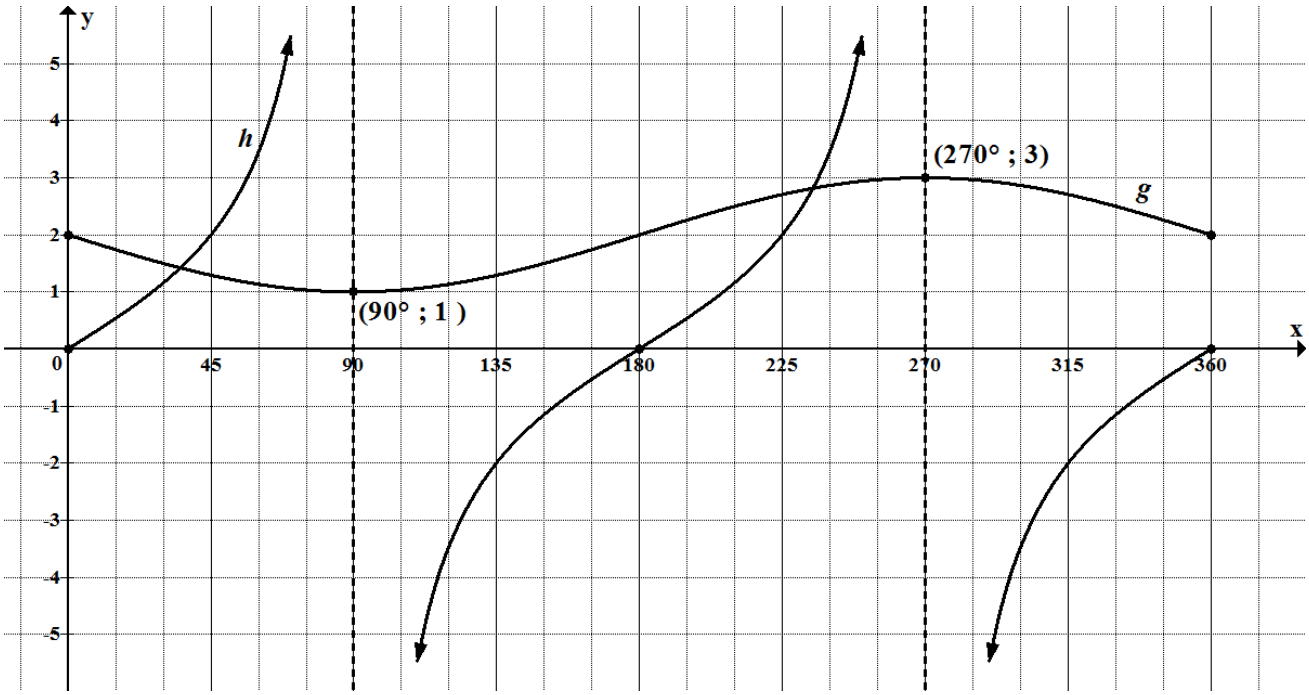
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[9]

**Question 10**

Sketched below are the graphs of $h(x) = 2tan\,x$ and $g(x) = b\,sin\,x + c$ for $x \in [0°; 360°]$.




(a) Determine the values of *b* and *c*.

_____ (2)

(b) Write down the period of *h*. _____ (1)

(c) Write down the equation of one of the asymptotes of *h*. _____ (1)

(d) Write down one value of *x* for which $g(x) - h(x) = 2$. _____ (1)

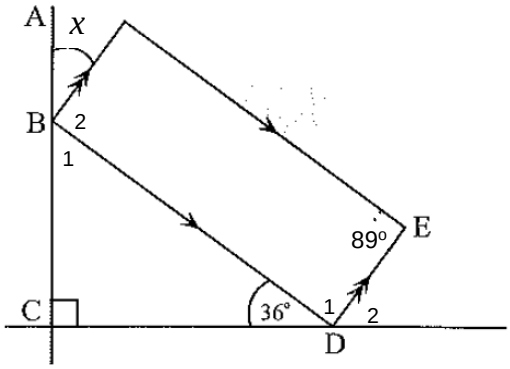
(e) Use the graph to determine the values of $x$ for which $h(x) \geq 2$

_____ (4)

[9]

**Question 12**

a) Calculate, with reasons, the size of the angle marked $x$.




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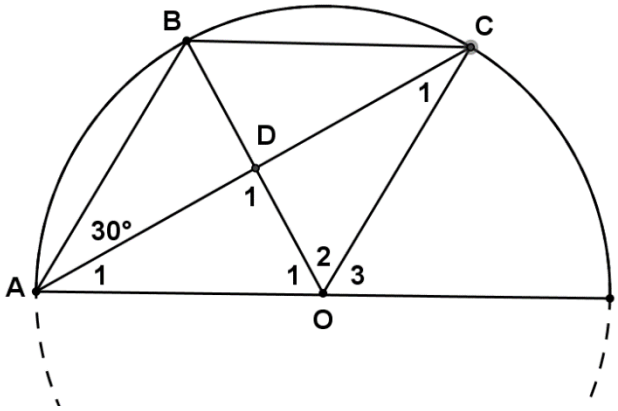
(3)

b) In the diagram below, O is the centre of the semi-circle.

A, B and C are on the circumference of the semi-circle.

ABCO is a **RHOMBUS**.

OB and AC intersect at D.




Determine, with reasons, the size of:

1) $\hat{C}_1$ _____

_____ (2)

2) $\hat{D}_1$ _____

_____ (2)

3) $\hat{O}_2$ _____

_____ (2)

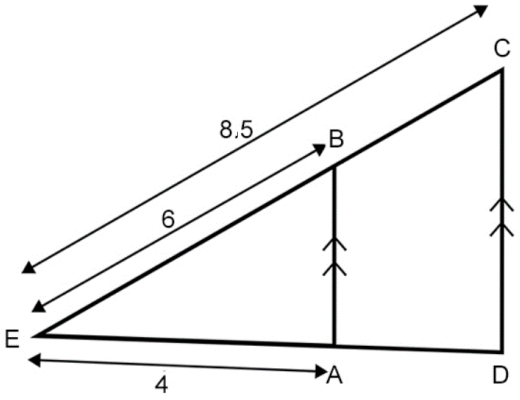
[9]

**SECTION D**

**NAME:**_____ **MATHS TEACHER:**_____

**Question 13**

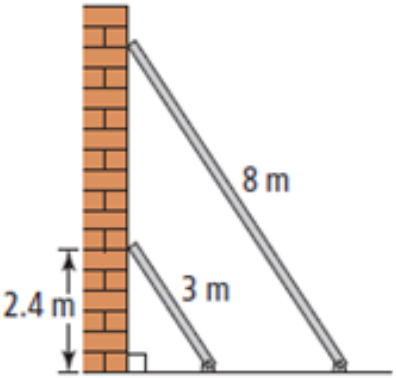
a) In $\Delta CED, AB // DC;\ EC = 8{,}5\ cm;$
$EB = 6\ cm\ ;\ EA = 4\ cm$
Determine the length of $AD$.




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(4)

b) Two extension ladders are leaning at the same angle, against a vertical wall.
The 3 metre ladder reaches 2,4 metres up the wall.




1) Prove that the two right-angled triangles are similar.
You will need to label the diagram. (2)

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2) How much further up the wall does the 8 metre ladder reach? Show clear reasoning.

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(4)

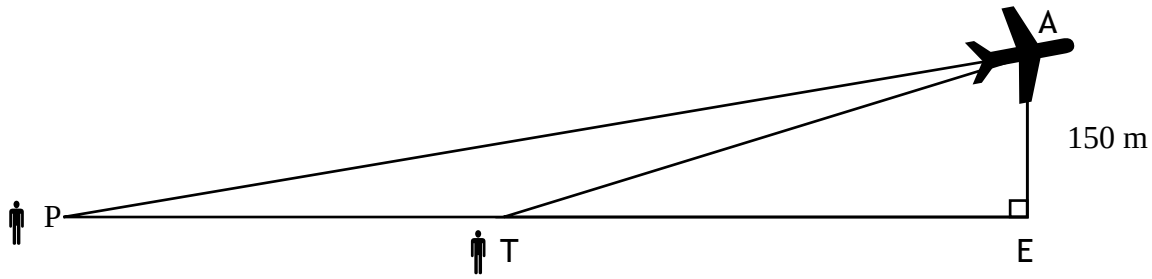
[10]

**Question 14**

Peter (P) and Tom (T) are observing an aeroplane in flight. From where Peter is standing he observes that the angle of elevation is 35°. Tom (T) notes that the aeroplane's angle of elevation was 40° from where he is standing.

At position E, the altitude of the plane was 150 m above the ground.

How far apart (length PT) are the two boys standing?




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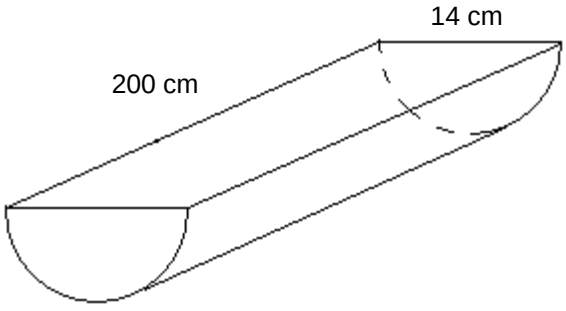
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[5]

**Question 15**

The diagram represents a water trough.

The length of the trough is 200 cm and the width at the top is 14 cm.




a) Calculate the volume of water that the trough can hold. (3)

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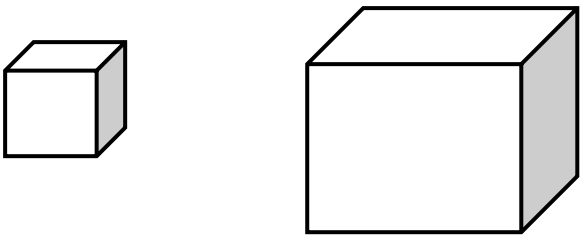
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b) A cube has a volume of 64 cm$^3$. If each side is scaled up by a factor of $x$, determine the volume of the bigger cube in terms of $x$. (2)

(Diagrams are not drawn to scale!)



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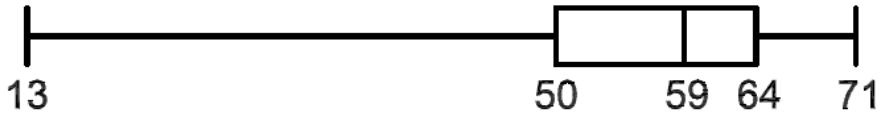
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[5]

**Question 16**

A set of data has a mean value of 52,1 units. A box and whisker diagram has been drawn based on this data set:




a) Is the data symmetrical, positively skewed or negatively skewed? (1)

_____

_____

b) If each data value in the set is multiplied by 3 and then has 5 added to it, determine which one of the following statements would be true for the new data set:

Circle your choice.

A $\bar{x} = 161,3$ and IQR $= 47$

B $\bar{x} = 156,3$ and IQR $= 42$

C $\bar{x} = 161,3$ and IQR $= 42$

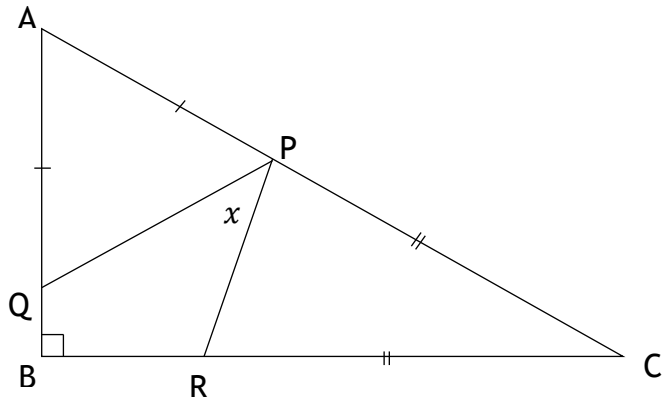
D $\bar{x} = 156,3$ and IQR $= 14$

(2)
[3]

**Question 17**

In $\Delta ABC$, points $P$, $Q$, and $R$ are chosen so that $AP = AQ$ and $CP = CR$.

Determine the value of $x$.




[5]